



Gamma Radiation Shielding and Thermal Performance of Concrete with Coconut Shell Ash Replacement

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Abstract

This research assessed the influence of replacing cement with coconut shell ash (CSA) on radiation shielding ability of a concrete using Nist XCOM, Phy-X, and Geant4. Four concrete C1, C2, C3, and C4 were prepared with water-to-cement (W/C) ratio of 0.6 throughout the mixes. The densities of the resulting concretes were altered (2.40 g/cm³ to 2.20 g/cm³) as the CSA content raises from 0.00 kg to 0.15 kg. More so, adding CSA to the concrete matrix, resulted in variation of linear attenuation coefficients (LAC) (0.195 cm⁻¹ for C1, 0.18 cm⁻¹ for C2, 0.183 cm⁻¹ for C3, and 0.177 cm⁻¹ for C4) of the resulting composite at 0.662 MeV. Based on this study, it was observed that samples C4 reduces the incoming photons better than C2, C3, and C1 at lower energy level (0.015 MeV to 0.04 MeV), while samples C1 reduces the incoming photons better than C2, C3, and C4 at higher energy level (0.04 MeV to 15 MeV). The thermal performance of the resulting concrete C2, C3, and C1 were improved when compared with sample C4. Therefore, concrete samples C2, C3, and C1 is the most effective for radiation shielding for their resistance to thermal threats.

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1.0 Introduction

Nuclear innovation is considered as the alternate source of energy which when utilized could mitigate the unexpected power issues [1]. This energy coming from nuclear technology gives out ionizing rays which could be very dangerous to human health [2]. It is for this reason that shielding becomes necessary in order to minimize the level of these rays

coming from their source to the nearest minimal, so that the population leaving close to the facilities that emits such hazardous radiation can be protected from the dangers of the rays [3, 4]. The massless and chargeless nature of gamma rays made them highly penetrative, necessitating the need to search for high proton number (Z) and density materials in order to achieve a better shielding against the harmful radiation [5].

On that note, researchers suggested ceramics, lead, metal alloys and Clay for gamma radiation shielding because of their high Z, high densities [6, 7]. But the limitations like frigidity for ceramics, low mechanical strength for Clay, health risks for metal alloys, and high cost as well as toxicity for lead, calls for a serious concern [8]. Concretes has been the substitute for these materials because of their high density and easy to operate, which makes them good for gamma radiation shielding [9]. On the quest to improve the shielding ability of concretes, researchers suggested that, concrete should be mixed with local wastes since some of these local wastes are high Z materials. Therefore, when local wastes like coconut shell powder is mixed with concrete, because of the silicon in it, the shielding ability of the resulting composite is expected to be altered, and also, because of the calcium content in it, the thermal resistance of the resulting concrete could be improved [10].

Therefore, it is vital to explore the addition of local content into concrete, and also to check the parameters like differential thermal analysis (DTA), which gives the extent of the material to withstand high degree of temperature generated by electromagnetic radiation before getting deformed [11-13].

Thermal performance of a material to be utilized for gamma radiation shielding is also very important to be assessed, as it gives the clear insight the amount of temperature (generated from electromagnetic radiation) that a material can withstand before getting deformed or cracked. It was reported by [14] that, some of these local materials like coconut shell ash (CSA), rice husk ash (RHA), bone ash (BA), palm nut shell ash (PNSA), and the rest, can improve the thermal resistance of a material due to the presence of calcium in them [15]. It is also crucial to check the linear attenuation coefficient (LAC, cm^{-1}) which describes the easiness of a material to shield radiation [16, 17].

By repurposing coconut shells, a typically discarded waste material, the utilization of CSA is viewed as a sustainable practice that can lead to a reduction in concrete production's environmental impact [18]. Moreover, since it is relatively cost-effective in comparison to alternative additives characterized by high density, CSA is a viable solution for increasing concrete's ability to shield against radiation from an economic perspective [19].

In this study, chemical content of the concrete-CSA is checked by the Energy Dispersive X-ray Fluorescence (EDXRF), and the data obtained are used to compute the GRSA for the resulting concrete-CSA material with the aid of Phy-X/PSD, Nist XCOM, and Grant4 software. The thermal characterization of the fabricated materials is also conducted in the work using DTA which depict the degree of temperature that a material can withstand before getting deformed or cracked.

2.0. Methodology

2.1. Production of Concrete - Coconut Shell Ash (CSA) Materials

Figure 1 depicted how concrete samples were prepared. C1, containing 0 wt% of CSA content replacing cement, was prepared by blending 0.5 kg of cement, 1 kg of sand, and 1 kg of granite. C2, with 10 wt% of CSA content as a substitute of cement, was prepared as follows: 0.45 kg of cement, 1 kg of sand, 1 kg of granite, and 0.05 kg of CSA. C3, having 20 wt% of the content of CSA as replacement of cement, was achieved by the mixture of 0.4 kg of cement, 1 kg of sand, 1 kg of granite, and 0.1 kg of CSA. C4, which has 30 wt% of CSA content replacing cement in it, was prepared with 0.35 kg of cement, 1 kg of sand, 1 kg of granite, and 0.15 kg of CSA. In this research work, we adopted 0.6 as the fraction of water-cement (W/C) for all mixtures [20, 21]. The resulting mixture was shaped like discs after a rigorous stirring, later which the molded concretes were spread under the sun for some days to regularize their weight. In order to make sure that the samples were free from the water content, the samples were put into an oven and the temperature of the oven was set to low degree for two hours.



Figure 1: Illustration of Fabrication Processes of Concrete-Coconut Shell Ash

The density (ρ , g/cm³) of the prepared samples were determined with the aid of Equation 1 based on the report of [22, 23] as presented in **Table 1**. The total masses (M, g) of the individual aggregates were recorded from a weighing balance (digital) which is as accurate as $\pm 1.0 \times 10^{-2}$ g. Then, the individual aggregate's volume (V, cm³) of used concretes were determined by pouring the individual sample powders into a beaker of water and measuring the change in the water level, which gives the volume of the particular aggregate. Summation of these individual volumes gives the total volume (V).

$$\rho \left(\frac{g}{cm^3} \right) = \frac{M}{V} \quad (1)$$

Table 1: Mixing Proportion of Concrete and Coconut Shell Ash

Sample ID	Sand (g)	Granite (g)	Cement (g)	CSA (g)	Density (g/cm ³)
A	1000.00	1000.00	500	0.00	2.40
B	1000.00	1000.00	450	50	2.28
C	1000.00	1000.00	400	100	2.24
D	1000.00	1000.00	350	150	2.20

2.2. Sample Analysis

The specimen's elemental composition was characterized through the analytical technique of energy-dispersive X-ray (EDX) spectroscopy, whereby a sample was bombarded by a targeted electron beam which displaced electrons from their atomic orbits, while distinctive X-rays are emitted. The bombardment of the sample with electrons causes ejection of electrons, with vacancies being created and populated by electrons of a lower energy, while as they relax excess energy in the X-ray form is subsequently released. During the period of relaxation, the energy given is absolutely distinct for each and every of the element within present withing the sample. In essence, each and every element possesses their own completely unique X-ray signature, as such, the X-rays emitted from the specimen furnished qualitative compositional insights [14, 24]. This qualitative compositions of the elements for the produced samples in wt% were carried out using ARL QUANT'X EDXRF Analyzer at Umaru Musa Yar'adua University (UMYU), Katsina, Nigeria, and the results are presented in Table 2.

The research conducted DTA test to measure the thermal transition of each of the samples through observation of temperature difference between the control material (Al₂O₃) and produced samples. Same quantity of both control and produced samples were respectively placed on the control and sample pans, and they were both exposed to same quantity of heat between room-temperature up to 200 °C. The graph of graph and sample temperature different was plotted as a function of temperature, and the endothermic and exothermic transition in the samples was obtained.

3.0. Results and Discussion

Table 2 provides insights into the elemental composition of the samples, revealing that the major elements in the samples are oxygen (O), silicon (Si), calcium (Ca), strontium (Sr), aluminum (Al), Fe and potassium (K), while the minor elements include Sulphur (S). Similar findings have been reported by [12, 16, 21].

Table 2: Qualitative Elemental Composition of the Fabricated samples in wt%

Elements	C1	C2	C3	C4
Fe	1.16235	1.34722	1.44845	2.10801
O	41.0922	40.8407	40.1273	39.1641
Si	18.9538	20.4865	21.0052	21.4102

Al	5.69045	4.02491	2.76121	1.10823
Mg	5.87318	3.70901	2.10012	0.36555
S	0.57845	0.66876	0.72641	0.81601
Ca	19.8359	21.0657	22.0168	22.8282
K	1.88879	2.43303	2.99102	3.47862
Sr	4.92441	5.42402	6.82344	8.72101

In terms of the most prominent element in the sample there is the emergence of oxygen (O), with 41.0923 (C1) and 39.1643 (C4) as the maximum and minimum wt%, respectively. This is followed by calcium (Ca) ranking second with 22.8283 (C4) and 19.8360 (C1) as the maximum and minimum wt%, respectively. Silicon (Si) follows as the second most abundant element, peaking at 21.4101 wt% (C4) and dropping to 18.9539 (C1). Magnesium (Mg) is also a significant element, reaching a high of 5.8732 wt% (C1) and falling to 0.3656 (C4). Aluminum (Al) is another notable element, with the highest wt% (C1) at 5.6905 and the lowest at 1.1082 (C4). Strontium (Sr) comes next, with a maximum wt% of 8.7210 (C4) and a minimum of 4.9244 (C1). Potassium (K) follows with a high of 3.4786 wt% (C4) and a low of 1.8888 (C1), and Iron (Fe) is seen with the highest value of 2.1080 wt% (C4) and the lowest at 1.1624 wt% (C1). Sulfur (S) is the least abundant element, peaking at 0.8160 wt% (C4) and dropping to 0.5785 wt% (C3).

In this study, the results of the elemental composition from Table 2, as well as the densities presented in Table 1 were then utilized to compute the parameters like LAC, which is the fundamental required for the analysis of the gamma radiation shielding ability (GRSA) of a material. These computations were carried out using the Phy-X/PSD software [25, 31] and Geant4 simulation [5, 25]. The results of two software were compared with Nist XCOM online software [25], all at the gamma energy of 0.015 MeV to 15 MeV.

The values of LAC of different prepared concrete at different energies ranging from 0.15 MeV to 15 MeV were calculated by Geant4 simulation, Phy-X/PSD and XCOM program and the relative deviation has been estimated according the Equation 2 and 3

$$Dev_1 (\%) = \frac{\mu_{XCOM} - \mu_{Phy-x}}{\mu_{Phy-x}} \times 100 \quad (2)$$

$$Dev_2 (\%) = \frac{\mu_{Geant4} - \mu_{Phy-x}}{\mu_{Phy-x}} \times 100 \quad (3)$$

The maximum deviation determined through the three different techniques was no more than 3% for the four samples studied in the energies range from 0.15 MeV to 15 MeV as shown in Table 3. at very low energy, specially at 0.015 MeV, the LAC increased with increasing the CSA. By increasing the energy, the LAC decreased slightly with increasing the CSA.

The LAC of the fabricated concretes at 0.015 MeV to 15 MeV gamma energies for all samples are plotted and presented in **Figure 2**. It could be observed at each energy that the LAC of the entire samples decreased as the energy is appreciating. As in literature, LAC depicts the fraction of a gamma rays beam at a given energy that is diffused or soak up per unit fatness of the absorber. From **Figure 2**, a slight decrease in the values of LAC (25.555 cm⁻¹ to 22.049 cm⁻¹ for C1, 28.956 cm⁻¹ to 21.098 cm⁻¹ for C2, 28.067 cm⁻¹ to 19.105 cm⁻¹ for C3, and 30.482 cm⁻¹ to 24.008 cm⁻¹ for C4) was observed between the lower energy range of 0.015 MeV to 0.02 MeV.

Table. 3. LAC of concrete samples at different energies using Geant4, Phy-X and XCOM.

Energy, MeV	C1					C2				
	LAC, XCOM	LAC, Phy-X	Dev1	LAC, Geant4	Dev2	LAC, XCOM	LAC, Phy-X	Dev1	LAC, Geant4	Dev2
0.015	25.128	25.555	-1.67	25.272	-1.11	29.480	28.956	1.81	28.835	-0.42
0.02	21.689	22.049	-1.63	21.773	-1.25	21.475	21.098	1.79	20.559	-2.55
0.03	7.238	7.358	-1.62	7.193	-2.24	7.089	6.965	1.78	6.785	-2.58
0.04	3.413	3.469	-1.63	3.402	-1.94	3.320	3.260	1.82	3.182	-2.40
0.05	1.979	2.012	-1.63	1.970	-2.10	1.915	1.881	1.79	1.841	-2.12
0.06	1.321	1.343	-1.63	1.294	-3.66	1.273	1.251	1.78	1.229	-1.76
0.08	0.774	0.787	-1.63	0.783	-0.45	0.742	0.729	1.78	0.722	-1.01

0.1	0.564	0.573	-1.65	0.579	1.05	0.539	0.530	1.78	0.528	-0.38
0.15	0.383	0.389	-1.65	0.386	-0.87	0.365	0.359	1.80	0.361	0.50
0.2	0.319	0.325	-1.64	0.321	-1.08	0.305	0.299	1.75	0.302	0.83
0.3	0.262	0.266	-1.59	0.261	-1.82	0.249	0.245	1.74	0.248	1.07
0.4	0.230	0.234	-1.64	0.229	-2.08	0.219	0.216	1.78	0.218	1.15
0.5	0.208	0.212	-1.63	0.212	-0.17	0.199	0.195	1.78	0.198	1.18
0.6	0.192	0.195	-1.63	0.198	1.42	0.183	0.180	1.78	0.182	1.20
0.8	0.168	0.171	-1.63	0.168	-1.66	0.160	0.158	1.78	0.159	1.21
1.0	0.151	0.153	-1.63	0.149	-2.79	0.144	0.141	1.78	0.143	1.22
1.5	0.123	0.125	-1.63	0.121	-2.86	0.117	0.115	1.79	0.116	1.23
2.0	0.106	0.108	-1.63	0.106	-1.56	0.101	0.100	1.77	0.101	1.21
3.0	0.087	0.089	-1.64	0.088	-1.10	0.083	0.082	1.77	0.083	1.16
4.0	0.077	0.078	-1.63	0.072	-7.28	0.074	0.072	1.79	0.073	1.11
5.0	0.070	0.071	-1.63	0.070	-1.68	0.067	0.066	1.80	0.067	1.06
6.0	0.066	0.067	-1.65	0.065	-2.58	0.063	0.062	1.79	0.063	1.01
8.0	0.061	0.062	-1.64	0.059	-3.93	0.059	0.058	1.78	0.058	0.91
10.0	0.058	0.059	-1.65	0.057	-3.52	0.056	0.055	1.78	0.056	0.83
15.0	0.055	0.056	-1.65	0.056	-0.55	0.054	0.053	1.80	0.053	0.70
Energy, MeV	C3					C4				
	LAC, XCOM	LAC, Phy-X	Dev1	LAC, Geant4	Dev2	LAC, XCOM	LAC, Phy-X	Dev1	LAC, Geant4	Dev2
0.015	27.574	28.067	-1.75	28.708	2.28	30.492	30.482	0.03	29.680	-2.63
0.02	18.769	19.105	-1.76	19.627	2.73	23.914	24.008	-0.39	24.309	1.25
0.03	6.191	6.302	-1.76	6.197	-1.66	7.900	7.899	0.02	7.667	-2.93
0.04	2.914	2.966	-1.75	3.086	4.03	3.676	3.676	0.00	3.640	-1.00
0.05	1.697	1.728	-1.76	1.741	0.79	2.099	2.099	0.01	2.052	-2.24
0.06	1.142	1.163	-1.75	1.177	1.28	1.378	1.378	0.01	1.374	-0.29
0.08	0.682	0.694	-1.76	0.682	-1.79	0.781	0.781	0.00	0.752	-3.69
0.1	0.506	0.515	-1.74	0.525	1.98	0.554	0.554	0.00	0.541	-2.38
0.15	0.352	0.358	-1.75	0.357	-0.16	0.363	0.363	0.01	0.361	-0.42
0.2	0.296	0.302	-1.74	0.299	-0.97	0.298	0.298	0.01	0.299	0.40
0.3	0.244	0.249	-1.75	0.245	-1.52	0.242	0.242	0.01	0.244	0.96
0.4	0.215	0.219	-1.76	0.216	-1.69	0.212	0.212	0.01	0.214	1.14
0.5	0.195	0.199	-1.75	0.195	-1.76	0.192	0.192	0.01	0.194	1.22
0.6	0.180	0.183	-1.75	0.180	-1.79	0.177	0.177	0.01	0.179	1.26
0.8	0.158	0.160	-1.75	0.158	-1.83	0.154	0.154	0.00	0.156	1.29
1.0	0.142	0.144	-1.75	0.141	-1.84	0.139	0.139	0.02	0.140	1.31
1.5	0.115	0.117	-1.76	0.115	-1.85	0.113	0.113	0.02	0.114	1.32
2.0	0.100	0.101	-1.76	0.100	-1.82	0.098	0.098	0.01	0.099	1.28
3.0	0.082	0.083	-1.75	0.082	-1.71	0.080	0.080	0.01	0.081	1.17
4.0	0.072	0.073	-1.76	0.072	-1.59	0.071	0.071	0.00	0.072	1.04
5.0	0.066	0.067	-1.75	0.066	-1.47	0.065	0.065	-0.01	0.066	0.90
6.0	0.062	0.063	-1.74	0.062	-1.35	0.062	0.062	0.02	0.062	0.79
8.0	0.057	0.058	-1.76	0.058	-1.14	0.057	0.057	0.01	0.058	0.57
10.0	0.055	0.056	-1.76	0.055	-0.97	0.055	0.055	-0.01	0.055	0.38
15.0	0.052	0.053	-1.76	0.053	-0.65	0.053	0.053	0.01	0.053	0.07

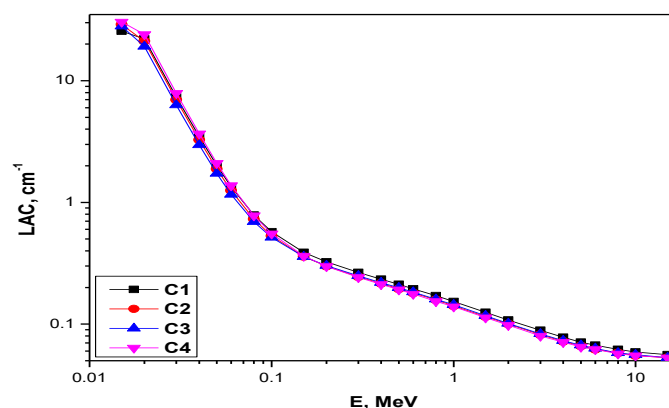


Figure 2: Linear attenuation coefficient (LAC) of the Fabricated Concretes against Energy

A fast drop in the values of LAC (22.049 cm^{-1} to 7.358 cm^{-1} for C1, 21.098 cm^{-1} to 6.965 cm^{-1} for C2, 19.105 cm^{-1} to 6.302 cm^{-1} for C3, and 24.008 cm^{-1} to 7.899 cm^{-1} for C4) was observed between the lower energy range of 0.02 MeV to 0.03 MeV. Between the energy range of 0.03 MeV to 15 MeV, a smooth decrease in the values of LAC was observed from 7.358 cm^{-1} to 0.056 cm^{-1} for C1, 6.965 cm^{-1} to 0.053 cm^{-1} for C2, 6.302 cm^{-1} to 0.053 cm^{-1} for C3, and 7.899 cm^{-1} to 0.053 cm^{-1} for C4. The overall reduction in the values of the LAC of the studied samples could be attributed to the decrease in the number of interactions at higher energy, confirming that, the shielding ability of the fabricated concretes is less at higher energy. On the other hand, at lower energy level, say 0.015 MeV, the values of the LAC of the studied concretes sample tend to slightly increase (25.555 cm^{-1} , for C1, 28.956 cm^{-1} for C2, 28.067 cm^{-1} for C3, and 30.482 cm^{-1} for C4) as the content of CSA is added to the concrete matrix. This increased in the values of LAC of concrete C4, C3, and C2 compared to C1 could be attributed to the increase in the concentration by Wt.% of silicon, Sulphur, calcium, potassium, and strontium in concrete C4, C3, and C2 compared to C1, confirming that, at lower energy, concrete C4 has the better gamma radiation shielding ability. Contrarily, at higher energy (5 MeV), the LAC of the studied concretes samples tend to slightly decrease as the content of CSA is added to the concrete matrix. This decrease for example 0.071 cm^{-1} for C1, 0.066 cm^{-1} for C2, 0.067 cm^{-1} for C3, and 0.065 cm^{-1} for C4 could be related to the high penetration of gamma radiation at higher energy, which resulted in a smaller number of interactions, confirming that, at higher energy level, concrete C1 has the better gamma radiation shielding ability. This result is in line with the report of [8].

The LAC at lower energy between 0.015 MeV to 0.04 MeV, raises as the content of CSA was slowly added to the concrete matrix. Sample C4 which has 150 g of CSA replacing cement consistently exhibited the highest LAC values at lower energies, making 150 g of CSA replacement a threshold for optimum concrete mix in radiation shielding application. The raise in LAC within this energy interval could be connected to the continual increase in the high atomic number (Z) elements like Strontium (Sr), Calcium (Ca), and Silicon (Si) concentrations with increase in the CSA content from 0 g to 150 g across the concrete samples as presented in Table 2. Their high Z could be responsible for the raise in the overall effective atomic number (Z_{eff}) of the concretes, resulting in an increase in density, being the major factor responsible for the improved LAC of our samples [15]. The superior LAC experienced at these energy intervals could also be related to the fact that in the lower energy region, the possibility of interactions is high, which could result in high absorption of the incoming photon radiation [14]. On the contrary, at higher energy, the LAC was observed to have decreased with raise in the content of CSA across the concrete matrix. This reduction in the LAC values could be attributed to the fact that, the probability of interaction reduces as the energy increase [25].

The derived weight loss curve is generally more sensitive to subtle differences in weight loss than the DTA curve and is therefore most commonly used for comparing similar materials. **Figure 3** compares the curves derived for samples C1, C2, C3 and C4 by traditional DTA (steady rate of heating), where a minimum variation was observed. The weight loss in C2 was observed to be the least according to the figure, with sample C4 having the greatest loss of weight, which could be attributed to the calcium carbonate decomposition and the CO_2 gas evaporation. The temperatures obtained are similar to those presented by [2, 26]. Each of such temperature of dehydroxylation is sufficient enough to offer each sample an ability of withstanding the possible internal heat which could be generated from gamma rays. Based on the results, sample C2 could withstand the heat generated by gamma radiation better than sample C3, then sample C1, and sample C4. This result is in line with the report of [2, 14].

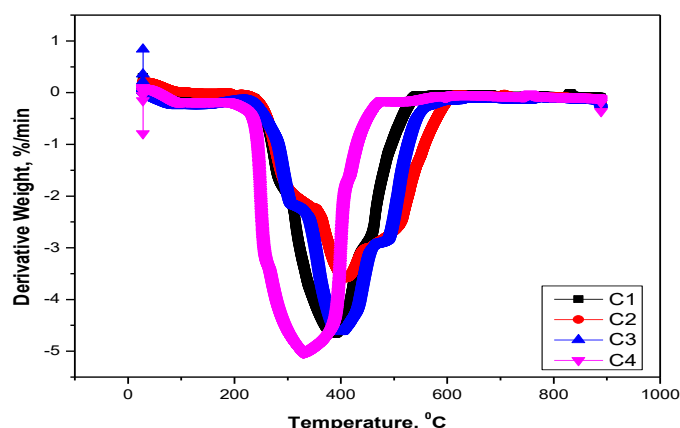


Figure 3: Differential Thermal Analysis of the Fabricated Concretes against Temperature

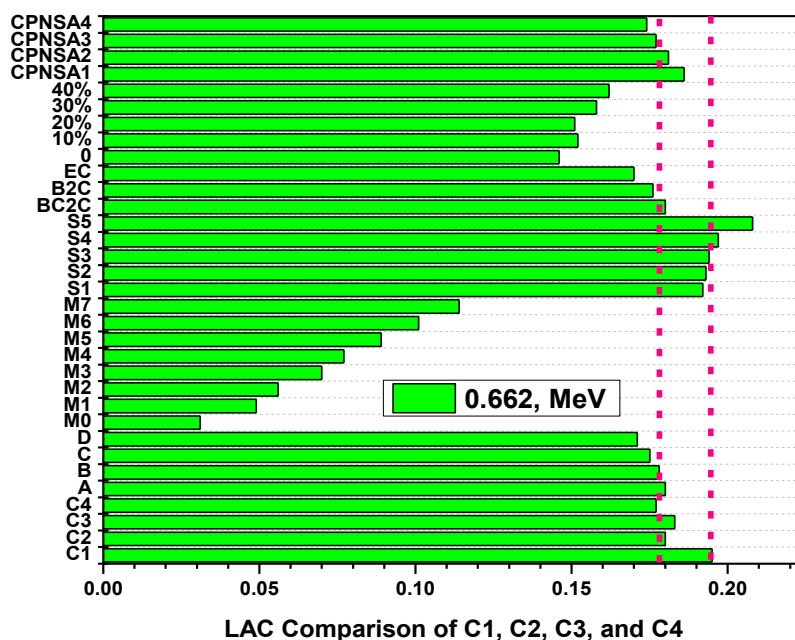


Fig. 4: Comparison of LAC for the developed concrete with some recently published concrete.

To validate the radiation shielding characteristics of our sample, we compared the LAC values in the present study with some lately published related concrete materials. Based on Figure 4, S4 and S5 recently published by [27] exhibited the superior LAC values respectively of 0.197 cm^{-1} and 0.207 cm^{-1} . The LAC of CSA- doped concretes at 0.662 MeV are 0.195 , 0.18 , 0.183 , and 0.177 cm^{-1} for samples C1, C2, C3, and C4, respectively. The mentioned LAC values for the developed concretes are close to that reported previously for samples B2C and BC2C [28] with LAC values of 0.175 and 0.178 cm^{-1} , respectively. Developed concrete samples C1, C2, C3, and C4 have LAC values close to those of concretes B2C and BC2C, which contain a high content of BaSO_4 and BaCO_3 . On contrast, fabricated concrete samples have LAC values higher than those reported for S1 ($\text{LAC} = 0.192 \text{ cm}^{-1}$), S2 ($\text{LAC} = 0.93 \text{ cm}^{-1}$), S3 ($\text{LAC} = 0.194 \text{ cm}^{-1}$) by [27], also has higher values compared to those reported for 0% ($\text{LAC} = 0.146 \text{ cm}^{-1}$), 10% ($\text{LAC} = 0.151 \text{ cm}^{-1}$), 20% ($\text{LAC} = 0.150 \text{ cm}^{-1}$), 30% ($\text{LAC} = 0.156 \text{ cm}^{-1}$), 40% ($\text{LAC} = 0.160 \text{ cm}^{-1}$), EC ($\text{LAC} = 0.167 \text{ cm}^{-1}$), M0 ($\text{LAC} = 0.030 \text{ cm}^{-1}$), M1 ($\text{LAC} = 0.048 \text{ cm}^{-1}$), M2 ($\text{LAC} = 0.054 \text{ cm}^{-1}$), M3 ($\text{LAC} = 0.069 \text{ cm}^{-1}$), M4 ($\text{LAC} = 0.076 \text{ cm}^{-1}$), M5 ($\text{LAC} = 0.089$

cm^{-1}), M6 ($\text{LAC} = 0.102 \text{ cm}^{-1}$), and M7 ($\text{LAC} = 0.113 \text{ cm}^{-1}$) [29, 30]. The comparison reveals that the produced concretes have an excellent radiation shielding capability when compared to previously reported concretes. As a result, it is suitable for radiation shielding applications in nuclear power plants and building construction.

4.0. Conclusion

At lower energy range (0.015 MeV to 0.04 MeV), it was observed that the composite samples C4 is better attenuating than concrete C2, C3, and C1 in shielding gamma rays, making 150 g of CSA replacement a threshold for optimum concrete mix in radiation shielding application at lower energy range between 0.015 MeV to 0.04 MeV. While at medium and higher energy range (0.06 MeV to 0.15 MeV), it was observed that the composite samples C1 is better attenuating than concrete C2, C3, and C4, making 0 g of CSA replacement a threshold for optimum concrete mix in radiation shielding application at higher energy interval of 0.06 MeV to 0.15 MeV. The introduction of CSA into the concrete matrix increases the thermal performance of the concrete as they were to withstand high degree heat generated by electromagnetic (EM) radiation before getting deformed, confirming that, the samples with CSA content can be applied in shielding of gamma radiation. Based on the findings of this study, the radiation subjections by staff during procedures in the nuclear medicine rooms could be greatly minimize by application of concretes mixed with CSA as shielding material. The application of concrete incorporated with CSA in shielding in a non-destructive testing (NDT) laboratory, safety could be assured, since the radiation coming from inspection equipment can be blocked, particularly at lower energies. Our materials could be utilized in the nuclear power plants as structural compositions in order to safeguard the radiation workers as well as the environment from ionizing effects of radiation, donating greatly to the operational safety and management of wastes. The high thermal resistance of our samples could be a great advantage to building industries, especially where high temperature structural buildings are required.

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